

TRAN VAN MANH

Robotics Engineer | Autonomous Navigation & Deep Reinforcement Learning

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PROFESSIONAL SUMMARY

I am a Robotics Engineer with more than 5 years of experience in autonomous navigation systems, mobile robots, and AI-driven robotic applications. I enjoy exploring new technologies, tackling challenging engineering problems, and transforming research ideas into real-world robotic solutions. Skilled in designing and deploying ROS/ROS2-based systems with behavior-tree architectures, path planning, with a proven record of transferring AI-based navigation policies from simulation to real-world platforms.

TECHNICAL SKILLS

Core Expertise: Autonomous Navigation, Robot Control Systems, Path & Motion Planning, Deep Reinforcement Learning, Behavior-Tree Architectures, Multi-Robot Systems, Sim-to-Real Transfer

Programming Languages: C++, C, Python, MATLAB

Robotics Frameworks: ROS / ROS2, Navigation Stack, AMCL, OpenCV, RTAB-Map, Kalman Filter

Simulation & ML Tools: Gazebo, Stage, PyTorch, Gym, NumPy, Matplotlib, LaTeX, GitHub

Soft Skills: Problem solving, teamwork, leadership

PROFESSIONAL EXPERIENCE

Robotics Engineer

Robot Global Team (RGT)

Sep 2024 – Present

Seoul, South Korea

- Developed and optimized local planning algorithms (sampling- and search-based methods) for autonomous navigation, integrating them successfully into the service robot architecture.
- Designed and implemented robot behaviors using Behavior Trees to handle complex, real-world deployment scenarios.
- Enhanced overall system performance by reducing computation costs on Single Board Computers (SBCs) by 30%, while ensuring smooth driving and control for both single- and multi-robot frameworks.
- Developed and maintained motor controller systems, including critical safety features (emergency button, voltage filtering, and battery logging) and stable serial communication.

Researcher – Intelligent Robot Laboratory

Chungbuk National University

Mar 2022 – Aug 2024

Cheongju, South Korea

ATC Project: Development and implementation of a navigation algorithm with collision avoidance based on deep reinforcement learning (Deep-RL).

- Successfully transferred deep reinforcement learning models from simulation training to real-world service humanoid robots.
- Evaluated mapless navigation and localization using a motion capture system, achieving over 95% navigation success rate and localization error below 10%, certified by KR.

Researcher

Vietnam–Korea Institute of Science and Technology (VKIST)

Aug 2020 – Feb 2022

Hanoi, Vietnam

Research and development of autonomous mobile robot technology for industrial transport applications.

- Defined functions, requirements, specifications, and technical criteria for the prototype autonomous robot system.
- Developed and designed the kinematic model and autonomous navigation framework for an omnidirectional mobile robot prototype.
- Optimized magnetic flux density in a PMSM motor using JMAG.
- **Key frameworks and tools:** genetic algorithm optimization, navigation-stack framework, AMCL, and Kalman filter.

Research Intern

Vietnam Academy of Science and Technology (VAST)

Mar 2018 – Aug 2020

Hanoi, Vietnam

Project: “Smart Human-Form IVASTBot Robot Applied in Communication and Serving Humans” (Phase 01).

- Designed an intelligent trajectory-tracking system for an omnidirectional mobile robot.

- Developed backstepping control and sliding mode control algorithms in MATLAB.
- Deployed low-level control in C for embedded systems on omnidirectional mobile robots.

EDUCATION

Master of Science in Control and Robot Engineering

Chungbuk National University

Mar 2022 – Aug 2024

Cheongju, South Korea

- GPA: 3.69 / 4.0
- **Thesis:** *Autonomous Navigation in Complex Environments Based on Deep Reinforcement Learning Using Selective Policy Switching for Mobile Robots.*

Bachelor of Science in Control Engineering and Automation

Hanoi University of Science and Technology

Aug 2015 – Aug 2020

Hanoi, Vietnam

- GPA: 3.12 / 4.0
- Ranked 1st in Vietnam for Electrical and Electronic Engineering by QS World University Rankings.

PUBLICATIONS & PATENTS

Publications

- T. V. Manh et al., “Cooperative Deep Reinforcement Learning Policies for Autonomous Navigation in Complex Environments,” *IEEE Access*, vol. 12, pp. 101053–101065, 2024. **(First author)**
- “Mapless Navigation with Distributional Reinforcement Learning,” *Journal of Korea Robotics Society*, 2024. **(First author)**
- “Adaptive Control for Uncertain Model of Omni-directional Mobile Robot Based on Radial Basis Function Neural Network,” *International Journal of Control, Automation, and Systems*, 2021. (Co-author)
- “Autonomous Navigation for Omnidirectional Robot Based on Deep Reinforcement Learning,” *International Journal of Mechanical Engineering and Robotics Research*, 2020. (Co-author)
- “Nonlinear Backstepping-Sliding Mode Control of Electro-Hydraulic Systems,” *ICERA 2019 – International Conference on Engineering Research and Applications*. (Co-author)
- “Mapping and Navigation with Four-Wheeled Omnidirectional Mobile Robot Based on Robot Operating System,” *2019 International Conference on Mechatronics, Robotics and Systems Engineering (MoRSE)*. (Co-author)
- “Neural Network Based Adaptive Control of Web Transport System,” *2019 International Conference on System Science and Engineering (ICSSE)*. (Co-author)
- “Trajectory Tracking Control for Four-Wheeled Omni-directional Mobile Robot Using Backstepping Technique Aggregated with Sliding Mode Control,” *2019 First International Symposium on Instrumentation, Control, Artificial Intelligence, and Robotics (ICA-SYMP)*. (Co-author)

Patents

- V. M. Tran and G.-W. Kim, “Autonomous Navigation System in Complex Environments Based on Deep Reinforcement Learning Using Selective Policy Switching for a Mobile Robot,” KR Patent MP24-0134 / MP24-0141, 2024.

ACHIEVEMENTS

- Best Paper Award, Journal of Korea Robotics Society, Korea, 2024.
- Scholarship for four semesters of the Master’s program at Chungbuk National University, 2022–2024.
- Best Graduation Thesis Defense for Undergraduates, Hanoi University of Science and Technology, Vietnam, 2020.